

LESSON 6-8 • SESSION 2 • PREREQUISITE PRACTICE A • SKILL 1 OF 2

Find Factors of a Number

Builds on Grade 4 • 4.OA.B.4

TODAY'S LESSONYou will factor $18x + 24$ with a number that divides both 18 and 24.**YOU NEED TO**Find the common factors of two numbers, like 12 and 20. **You will use it in Practice Form A, problems 2 and 7.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** List all the **factors** of the first number.
- 2** List all the **factors** of the second number.
- 3** Find the numbers on **both** lists.

WATCH ME*the steps in action***Find the common factors of 12 and 20.****Step 1** 12: 1, 2, 3, 4, 6, 12**Step 2** 20: 1, 2, 4, 5, 10, 20**Step 3** On both lists: 1, 2, 4**Answer** **1, 2, 4****TRY IT WITH HELP***fill in each step*

- 1.** Find the common factors of 15 and 35.

Step 1 15: 1, _____, 5, 15**Step 2** 35: 1, 5, _____, 35**Step 3** On both lists: _____ and _____

- 2.** Find the common factors of 16 and 24.

Step 1 16: 1, 2, _____, 8, 16**Step 2** 24: 1, 2, 3, _____, 6, 8, 12, 24**Step 3** On both lists: 1, 2, _____, _____**YOUR TURN***use the steps on your own*

- 3.** List the common factors of each pair.

a. 12 and 18: _____**b.** 10 and 15: _____**c.** 14 and 21: _____**d.** 20 and 30: _____

Divide by a Common Factor

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TODAY'S LESSONYou will factor $30c + 42$ as $6(5c + 7)$.**YOU NEED TO**Divide two numbers by a common factor, like $30 \div 6 = 5$ and $42 \div 6 = 7$.**You will use it in Practice Form A, problems 2, 6, and 7.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Divide** the first number by the common factor.
- 2 Divide** the second number by the common factor.
- 3 Check:** multiply each answer by the factor.

WATCH ME*the steps in action***Divide 36 and 60 by 12.****Step 1** $36 \div 12 = 3$ **Step 2** $60 \div 12 = 5$ **Step 3** $12 \times 3 = 36$ and $12 \times 5 = 60$ **Answer 3 and 5****TRY IT WITH HELP***fill in each step***4.** Divide 18 and 24 by 6.**Step 1** $18 \div 6 = \underline{\hspace{2cm}}$ **Step 2** $24 \div 6 = \underline{\hspace{2cm}}$ **Step 3** $6 \times 3 = \underline{\hspace{2cm}}$ and $6 \times 4 = \underline{\hspace{2cm}}$ **5.** Divide 14 and 35 by 7.**Step 1** $14 \div 7 = \underline{\hspace{2cm}}$ **Step 2** $35 \div 7 = \underline{\hspace{2cm}}$ **Step 3** $7 \times 2 = \underline{\hspace{2cm}}$ and $7 \times 5 = \underline{\hspace{2cm}}$ **YOUR TURN***use the steps on your own***6.** Divide both numbers by the common factor.**a.** 12 and 20 by 4 $\underline{\hspace{2cm}}$ **b.** 15 and 35 by 5 $\underline{\hspace{2cm}}$ **c.** 20 and 45 by 5 $\underline{\hspace{2cm}}$ **d.** 16 and 40 by 8 $\underline{\hspace{2cm}}$ **e.** 24 and 36 by 12 $\underline{\hspace{2cm}}$ **f.** 10 and 25 by 5 $\underline{\hspace{2cm}}$ **READY CHECK***a problem like the one in today's lesson***7.** Which shows $18x + 12$ factored with the GCF?**A** $6(3x + 2)$ **B** $2(9x + 6)$ **C** $6(3x + 12)$ **D** $3(6x + 4)$ **CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about # $\underline{\hspace{1cm}}$